

Arduino LED Circuit Project Report

Introduction

This project involved building a basic electronic circuit using an **Arduino Uno** microcontroller board connected to a breadboard. The goal was to control two LEDs — one green and one red — using digital output pins on the Arduino. The circuit was wired on a standard breadboard, with resistors used to protect the LEDs from excess current. Power and control signals were supplied from the Arduino's digital I/O pins, and the board itself was connected to a computer via a USB cable for programming and power. This type of project is a foundational exercise in electronics and embedded systems, teaching the fundamentals of GPIO (General Purpose Input/Output) control, circuit wiring, and component handling.

Problems Faced

During the course of building this project, several challenges were encountered:

1. **Wiring Difficulties** Connecting the jumper wires correctly on the breadboard proved tricky at first. The internal connections of a breadboard are not visible, which made it easy to accidentally place components or wires in the wrong rows or columns, causing open circuits or short circuits.
 2. **LED Polarity and Orientation** LEDs are polarized components, meaning they only work when connected in the correct direction (anode to positive, cathode to ground). Identifying the longer leg (anode) versus the shorter leg (cathode) and placing them correctly required careful attention.
 3. **Resistor Placement** Ensuring that current-limiting resistors were placed correctly in series with each LED was another challenge. Misplacing a resistor or using the wrong resistance value could either prevent the LED from lighting up or risk burning it out.
 4. **Loose Component Connections** Breadboard holes can sometimes hold components loosely, leading to intermittent connections. Some wires and component legs needed to be reinserted firmly to ensure a stable electrical connection.
 5. **Arduino Pin Identification** Correctly identifying and using the right digital pins on the Arduino Uno for the LED outputs took some careful reference to the board's pin layout, especially when multiple wires were connected close together.
 6. **Code Upload and Debugging** Minor issues arose when uploading code from the computer to the Arduino via USB, including selecting the correct COM port and board type in the Arduino IDE.
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